26900355 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEMATA02 - DESIGN AND ANALYSIS OF ALGORITHMS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **5** questions

Each question carries **2** marks

1. Why is running time generally expressed in terms of input size rather than actual execution time in seconds? [U] / [CO1]
2. Define breadth-first search (BFS). [K] / [CO2]
3. What is meant by exploring a vertex during DFS? [K] / [CO2]
4. Among the intervals (1, 5), (2, 4), and (3, 6), which interval would be considered first by the earliest-finish-time-first strategy? [A] / [CO3]
5. Write the recurrence relation for an algorithm that divides an input of size n into three subproblems of size $n/3$ and performs n additional operations. [A] / [CO3]
6. What is meant by a representative element in a Union-Find data structure? [K] / [CO3]
7. Why must the subproblems be processed in a natural order from smaller to larger subproblems in the iterative approach? [U] / [CO4]
8. If $Z \leq_P Y \quad \text{and} \quad Y \leq_P X,$ [K] / [CO5]
   What conclusion can be drawn about the relationship between Z and X?

[2x5 = 10]

Part B

Short Essays/ Problems

Answer any **5** questions

Each question carries **6** marks

9. State and explain the property concerning the sum of functions. If $f(n) = O(h(n))$ and $g(n) = O(h(n))$, show that $f(n) + g(n) = O(h(n))$. [An] / [CO1]

10. Explain the scheduling-to-minimize-lateness problem using a real-life example involving project submissions, examination evaluation, or manufacturing orders. [U] / [CO3]

11. For items with weights $w_1 = 2$,$w_2 = 3$, and $w_3 = 4$, and capacity $W = 6$, use the dynamic programming recurrence to determine $\mathrm{OPT}(3, 6)$. Show the relevant computations. [A] / [CO4]

$$\mathrm{OPT}(i, w) = \max\{\mathrm{OPT}(i-1, w), w_i + \mathrm{OPT}(i-1, w-w_i)\}, \qquad w \geq w_i.$$

12. (a) Explain the two possible cases considered for an optimal solution O_j:$j \in O_j$ and $j \notin O_j$. [A] / [CO4]
    (b) For
    $$p(4) = 2, \qquad v_4 = 7, \qquad opt(2) = 5, opt(3) = 8,$$
    calculate $opt(4)$ and determine whether interval
    belongs to an optimal solution using the membership condition.

13. (a) State the running time required by the algorithm to compute all $e_{i,j}$ values and the subsequent dynamic-programming table. [An] / [CO4]
    (b) If there are $n = 100$ points, state the asymptotic number of segment-error computations and the asymptotic time required for the dynamic-programming stage.
    $$\text{Time for all } e_{i,j} = O(n^3), \qquad \text{Time for DP stage} = O(n^2).$$

14. (a) Define the Traveling Salesman Problem (TSP) in its decision form. [U] / [CO5]
    (b) For four cities v_1, v_2, v_3, v_4, a tour $v_1 \to v_2 \to v_3 \to v_4 \to v_1$ has distances $2, 3, 4, 5$, respectively. Determine its total length and decide whether it satisfies the bound $D = 15$.

15. (a)Define the Circuit Satisfiability Problem, [A] / [CO6]
    (b) For the Boolean circuit $K = x_1 \vee x_2$, give a satisfying assignment for its inputs.

16. (a) Define a certificate for a decision problem. [A] / [CO6]
    (b) For a 3-SAT instance, identify a suitable certificate for a “yes” instance and state what the certifier checks.

[6x5 = 30]

Part C

Essays

Answer any **3** questions

Each question carries **10** marks

17. Define a tree and discuss any four equivalent characterizations of a tree. For the graph $G = (V, E)$, where $V = \{1, 2, 3, 4, 5\}$ and $E = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$, determine whether G is a tree and justify your answer using two characterizations. [An] / [CO1]

18. Explain how three nested loops can lead to an $O(n^3)$ running time. Illustrate the analysis using the problem of determining whether two sets have an element in common. [An] / [CO2]

19. A transportation department wishes to connect five towns with minimum road-construction cost. [C] / [CO3]

    The possible roads are
    $AB = 8, \quad AC = 2, \quad BC = 3, \quad BD = 7, \quad CD = 4, \quad CE = 6, \quad DE = 5.$
    (i) Formulate the problem as a minimum spanning tree problem.
    (ii) Construct the minimum spanning tree.
    (iii) Determine the minimum total construction cost.
    (iv) Explain how the selected roads guarantee connectivity.

20. Mergesort divides a problem of size n into two subproblems of approximately equal size and spends linear time combining their solutions. For input sizes
    $8, \quad 16, \quad 32, \quad 64,$
    analyze the corresponding recursion structures. Determine the number of levels and explain how the amount of work changes from one level to the next. [An] / [CO3]

21. (a) State the noncrossing condition for an RNA secondary structure. [A] / [CO4]
    (b) Consider the pairs (1, 6) and (2, 5). Determine whether they satisfy the non-crossing condition.
    (c) Consider the pairs (1, 4) and (2, 5). State whether these pairs can occur together under the noncrossing condition.

22. 1.Define polynomial-time reducibility. What does $Y \leq_P X$ signify? [A] / [CO5]
    2.For a graph with $n = 8$ vertices, a Vertex Cover of size at most $k = 3$ is given as the decision requirement. Determine the corresponding Independent Set size that must be considered.

[10x3 = 30]